ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 0.005 mm	2 / 400 = 0.005 mm.
2	**(a)** has similar cells doing one job	Organs have more than one tissue type.
3	**(d)** provide contrast	Stains help visibility.
4	**(a)** a leaf	Only plant cells have all three.
5	**(d)** 10x	400 / 40 = 10.
6	**(c)** 10 mm	0.1 x 100 = 10 mm.
7	**(a)** an onion bulb	Underground bulb cells lack chloroplasts but have cell walls.
8	**(a)** They are large and store water and nutrients	Vacuoles in plants are large.
9	**(c)** nucleus and cytoplasm	Common parts.
10	**(b)** They have genetic material but no true nucleus.	Prokaryotic cells.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They lack a nucleus enclosed by a membrane (1); They are much simpler and smaller (1).	1+1
12	Prokaryotic: no true nucleus (bacteria) (1); Eukaryotic: true nucleus (plants, animals) (1).	1+1
13	Cytoplasm: jelly-like fluid with organelles (1); Nucleus: has genetic material, controls cell (1).	1+1
14	Total = 15 x 20 (1); = 300x (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Plant cell (1); Chloroplasts (1); Make food by photosynthesis (1).	1+1+1
16	Actual = 4 / 200 = 0.02 mm (1); 1 mm = 1000 micrometres (1); 0.02 mm = 20 micrometres (1).	1+1+1
17	A: nerve cell, sends messages (1); B: red blood cell, carries oxygen (1); C: muscle cell, produces movement (1).	1+1+1
	OR: Animal cell (1); Has nucleus and cytoplasm (1); No wall/chloroplast (1).	1+1+1
18	(a) Chloroplast (1); (b) Nucleus (1); (c) Vacuole (1).	1+1+1
	OR: Yeast: unicellular, bread (1); Mushroom: multicellular, food (1); Both lack chlorophyll (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plant: nucleus, wall, chloroplast, large vacuole (1); Animal: nucleus, no wall, no chloroplast, small vacuoles (1); Bacteria: no true nucleus, wall present, no chloroplast (1); Bacteria are the smallest (1); Bacteria have a nucleoid (1).	1+1+1+1+1
	OR: Non-cellular, extremely small (1); Multiply inside host cells (1); Diseases: influenza, dengue, measles (1); Prevention: vaccination (1); Hygiene and avoiding crowded places (1).	1+1+1+1+1
20	Parts: eyepiece, objective, stage, mirror (1); Light passes through the specimen (1); Lenses magnify the image (1); Total = 12 x 50 (1); = 600x (1).	1+1+1+1+1
	OR: Nucleus: absent vs present (1); Size: small vs large (1); Example: bacteria vs plants/animals (1); Complexity: simple vs complex (1); Nuclear membrane absent vs present (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.
- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** protozoan	Protozoa have a nucleus.
2	**(d)** contains genetic material	It is the genetic region.
3	**(c)** More space is available for carrying oxygen	More space for haemoglobin.
4	**(b)** producing chlorophyll	That is the role of the chloroplast.
5	**(d)** sugar	Yeast ferments sugars.
6	**(a)** Virus - unicellular	Viruses are non-cellular.
7	**(b)** pathogens	Disease-causing agents.
8	**(a)** absorption of water	Increases surface area.
9	**(a)** a fungus	From Penicillium mould.
10	**(c)** a tissue	Similar cells.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They enter a host cell (1); They use the host machinery to make copies (1).	1+1
12	Light can pass through it (1); A single layer shows cells clearly (1).	1+1
13	A plant cell has a rigid cell wall (1); Animal cells have only a flexible membrane (1).	1+1
14	Antibiotics act on bacterial structures (1); Viruses lack these and live inside host cells (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Body of multicellular organisms is large and complex (1); Division of labour is needed (1); A single cell does everything in unicellular ones (1).	1+1+1
16	Bodies are made of cells (1); Cells carry out life functions (1); Cells arise from pre-existing cells (1).	1+1+1
17	It helps in identifying microbes in samples (1); Doctors know the cause (1); Right treatment is given (1).	1+1+1
	OR: Bacteria are cells; viruses are not (1); Bacteria multiply by themselves; viruses need host cells (1); Bacteria are bigger than viruses (1).	1+1+1
18	Onion cell is rectangular with a clear cell wall (1); Cheek cell is rounded and irregular without a wall (1); Both have a nucleus and cytoplasm (1).	1+1+1
	OR: Keeps the specimen flat (1); Protects the objective lens (1); Prevents drying and air bubbles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cell: gland cell (1); Tissue: muscle tissue in the stomach wall (1); Organ: stomach (1); System: digestive system (1); Organism: human being (1).	1+1+1+1+1
	OR: Collect a drop of pond water on a slide (1); Put a cover slip (1); Observe under low and high power (1); Algae, Paramecium, Amoeba (1); Draw and label (1).	1+1+1+1+1
20	Decomposition of dead matter (1); Nitrogen fixation by Rhizobium (1); Curd and cheese making (1); Production of vitamins in the gut (1); Clean-up of waste (1).	1+1+1+1+1
	OR: Microbes were seen for the first time (1); Germs were linked to disease (1); Cell theory emerged (1); Hygiene and vaccines developed (1); New branches like microbiology started (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.
- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.

- Forgetting the stain, thin peel and cover slip when describing slide preparation.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** carrying signals over distance	Shape suits function.
2	**(c)** chloroplast	Amoeba does not make food by photosynthesis.
3	**(b)** virus, bacterium, red blood cell, human egg	Virus (about 0.1 micrometre) < bacterium (about 2 micrometres) < red blood cell (about 7 micrometres) < human egg (about 100 micrometres).
4	**(b)** moving organisms like Paramecium	Pond water has microscopic life.
5	**(a)** carbon dioxide and alcohol	It makes dough rise.
6	**(a)** rigid cell wall	The wall prevents bursting.
7	**(d)** chlorophyll	Fungi are not photosynthetic.
8	**(a)** influenza (viral)	Antibiotics act on bacteria.
9	**(a)** different organs for each function	A single cell has no organs.
10	**(b)** just inside the cell wall	Wall is the outermost layer.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The one cell carries out all functions (1); Food, gas exchange and excretion occur through the cell surface (1).	1+1
12	Provides more room for haemoglobin (1); So more oxygen is carried (1).	1+1
13	Leaves carry out photosynthesis (1); Roots are underground, lacking light (1).	1+1
14	Yeast ferments sugar in dough (1); Carbon dioxide bubbles make the dough rise (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Long thin projection (1); Increases surface area for absorption (1); Takes in water and minerals from soil (1).	1+1+1
16	Organisms that break down dead matter (1); Bacteria in soil (1); Moulds or mushrooms (1).	1+1+1
17	Bacterium has no true nucleus; amoeba has (1); Bacteria are far smaller (1); Amoeba is a protozoan (1).	1+1+1
	OR: Cells develop special shapes and structures for particular functions (1); Nerve cells: send signals (1); Muscle cells: contract (1).	1+1+1
18	Yeast ferments sugar (1); Releases carbon dioxide (1); Warmth speeds up yeast activity (1).	1+1+1
	OR: Host: organism in which another lives and feeds (1); Parasite: lives at the cost of the host (1); e.g. influenza virus in humans (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Single-celled fungus (1); Has nucleus and cell wall (1); Feeds on sugars (1); Releases carbon dioxide in bread dough (1); Produces alcohol in brewing (1).	1+1+1+1+1
	OR: Wall: plant cell only (1); Chloroplast: plant cell only (1); Vacuole is large in plant cells (1); Rigid wall gives a regular shape (1); Animal cells are flexible and irregular (1).	1+1+1+1+1
20	Nerve cell (1); Muscle cell (1); Red blood cell (1); Root hair cell (1); Guard cell or leaf cell (1).	1+1+1+1+1
	OR: Cause diseases: typhoid (1); Spoil food: mould on bread (1); Damage crops: rust fungus (1); Cause decay of wood (1); Prevention: hygiene, preservation (1).	1+1+1+1+1

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